Keshav Varshney

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WORK EXPERIENCE

•Software Engineer | *HPE* Bangalore, India | *Sept '24 - Present*

- **Hardware Management Service (HMS)**
  - Currently handling customer issues across **3 core HMS services** in Cray **CSM**: **SLS**, **FAS**, and **RTS**.
  - Troubleshoot **SLS (System Layout Service)** issues as the system-design source of truth for node/cabinet topology and network mappings.
  - Resolve **FAS** workflows and **RTS (Redfish Translation Service)** issues, which aid management of hardware not managed by Redfish (for example, ServerTech PDUs in River cabinets).
- **Internal Tool: ATLAS (Automated Triaging & Log Analysis Solution)**
  - Led a team to build a 3-stage AI RCA pipeline, reducing manual triage by **~80%**.
  - Built **CPV** with **50+ automated pre-upgrade checks** across Kubernetes, Ceph, PostgreSQL, etcd, & networks.
  - Built FAISS semantic search over **66+ docs** and **3000+** JIRA tickets with **zero** manual log correlation.
  - Integrated a TypeScript MCP server with VS Code Copilot Chat, exposing **14 tools** for structured RCA generation.
  - Rolled out ATLAS to **3+ teams**, delivering production diagnostics and reducing mean time to resolution (MTTR).
- **Rack Resiliency Service (RRS)**
  - Designed and implemented a backend service to manage critical services and zones for HPE HPC clusters.
  - Developed **CLI (cray rrs)** with **7 new commands** to list and describe zones and critical services, and create new service definitions through **JSON** payloads.
  - Mapped Kubernetes topology zones with Ceph storage zones, enabling a unified resiliency view and reducing manual intervention during service disruptions **by 30%**.
  - Containerized and deployed the backend as a pod with APIs exposed through a virtual service, improving accessibility and integration with other cluster tools.
  - Developed and maintained **20 automated API test cases** for RRS using Python, improving test coverage and reducing manual validation effort **by 40%**.
- **Platform Engineering Team**
  - Upgraded **cray-postgres-operator** from **v1.10.0** to **v1.14.0** for Cray CSM environments, validating compatibility and rollout behavior across operator-managed PostgreSQL components.
  - Implemented and validated upstream changes in *Cray-HPE/cray-postgres-operator* to execute a **4-minor-version** operator upgrade (**v1.10.0 to v1.14.0**), reducing **high CVEs to 0** and **medium CVEs by 300**.

•HPE Internship | *HPC Software Engineer* Bangalore, India | *Feb '24 - Aug '24*

- Upgraded **Istio** from **v1.11.8 to v1.19.10** across large Cray clusters with minimal downtime.
- Built automation to validate Istio versions across pod and container images (pilot and proxyv2).
- Implemented **rollout restart workflows** for non-compliant pods with stabilization and rollout checks.
- Added post-upgrade checks to improve reliability/observability and enforce correct Istio versions.
- Documented upgrade procedures, test cases, and known issues for future service-mesh upgrades.

•HPC Security Dashboard | *HPE-CTY Program* keshav-06/HPC-Security-Dashboard | *Jan '23 - Jun '23*

- Built a **Grafana** security dashboard to visualize **Kubernetes** cluster metrics.
- Integrated **Cilium** with Kubernetes to strengthen network-policy enforcement.
- Used Docker, Prometheus, and Kube-Bench for containerization, monitoring, and compliance checks.
- Gained hands-on exposure to Grafana, Prometheus, **eBPF**, and Cilium in K8s security workflows.

TECHNICAL SKILLS

- **Programming Skills:** C, C++, Python, Java, & SQL
- **Cloud Native:** Docker, Kubernetes, Istio, Helm, Ansible
- **Certification:** CKAD (Certified Kubernetes Application Developer)
- **API Development:** Flask, Flask-RESTful, Swagger
- **CI/CD:** GitHub Actions, Jenkins
- **Miscellaneous:** Linux Shell, unittest, mypy, pytest, pylint

EDUCATION

Year	Degree	Institute	CGPA/%
2020-2024	B.Tech., Computer Science and Engineering	Vellore Institute of Technology, Vellore	9.30/10
2019	Class 12th - CBSE Board	Silver Stone Public School, Chandausi	94.4%
2017	Class 10th - CBSE Board	R.R.K. Sr. Sec. School, Chandausi	10/10